

Autonomous Social Media Content Strategy Generator

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Course: B.Tech

Academic Year: 2025 - 2026

Project Type: AI-Based Web System

Institution: United College of Engineering and Research

Project Repository URL: <https://github.com/your-username/your-repository>

Demo / Prototype URL: <https://your-demo-link-here>

1. Introduction

The Autonomous Social Media Content Strategy Generator is an AI-powered system designed to assist businesses and content creators in managing their social media presence. It automatically generates content ideas, captions, hashtags, and posting schedules.

2. Problem Statement

Managing multiple social media platforms requires time, creativity, and marketing skills. Small businesses and individuals often struggle to maintain consistency and engagement.

3. Objectives

- 1 To automate social media content planning
- 2 To generate captions and hashtags using AI
- 3 To improve online engagement
- 4 To reduce manual workload
- 5 To provide an easy-to-use system

4. Literature Survey

Various digital marketing tools and AI-based platforms exist for social media management. However, most are expensive and complex for beginners. This project aims to provide a simple and cost-effective alternative.

5. Proposed System

The proposed system collects user inputs such as business type, target audience, platform, and tone. Based on this information, the AI module generates a structured content strategy.

6. System Architecture

The system architecture consists of four main components:

- 1 User Interface (Designed using Figma)
- 2 Input Processing Module
- 3 AI Content Generation Module
- 4 Output Display Module

7. Tools and Technologies Used

- 1 Figma - UI/UX Design
- 2 HTML, CSS, JavaScript - Frontend Development
- 3 AI Prompt Engineering
- 4 GitHub - Version Control
- 5 Markdown - Documentation

8. Implementation Methodology

- 1 Requirement Analysis
- 2 User Interface Design
- 3 Prompt Development
- 4 System Testing
- 5 Documentation Preparation

9. Features of the System

- 1 Automatic content planning
- 2 7-day content calendar generation
- 3 Caption and hashtag suggestions
- 4 Best posting time recommendation
- 5 Platform-specific strategies
- 6 User-friendly interface

10. Advantages

- 1 Saves time and effort
- 2 Improves brand visibility
- 3 Easy to operate
- 4 Cost-effective solution
- 5 Suitable for beginners

11. Limitations

- 1 Depends on AI accuracy
- 2 Requires stable internet connection
- 3 Limited customization
- 4 No auto-posting in current version

12. Use Case Description

A small business owner enters business details into the system. The system generates a weekly content plan along with captions and hashtags. The user posts the generated content on social media platforms.

13. Testing and Validation

The system was tested with different business profiles and target audiences. Generated outputs were evaluated based on relevance, creativity, and usability.

14. Results and Discussion

The system successfully generated meaningful content strategies. Users reported improved planning efficiency and better engagement.

15. Future Scope

- 1 Integration with social media APIs
- 2 Automatic post publishing
- 3 Advanced analytics dashboard
- 4 Mobile application development
- 5 Personalized AI models

16. Conclusion

This project demonstrates the effectiveness of AI in simplifying social media management. It provides an intelligent solution for content planning and branding.

17. References

- 1 OpenAI Documentation
- 2 Figma Community Resources
- 3 Digital Marketing Blogs
- 4 GitHub Documentation